

PRODUCT REQUIREMENTS DOCUMENT

Step Sync & Activity Challenge

Habit Health — Engagement Module

Product	Habit Health — Engagement Module
Feature	Step Sync & Activity Challenge
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1. Executive Summary

1.1 Problem Statement

Users need a seamless way to track their daily steps and compete with colleagues in friendly challenges. Current issues include confusion about connecting health apps (especially on Android where Health Connect requires a third-party tracker app), lack of engagement features around step tracking, and no social competition element to motivate users.

1.2 Solution Overview

- Improved step sync connection flow with clear guidance for Android and iOS users
- Activity Tracker widget showing real-time step count with multiple contextual states
- Activity Challenge page for viewing leaderboards and inviting colleagues
- Individual Challenge system for 1-on-1 or group competitions
- Streak tracking to encourage daily engagement
- Background sync every 15 minutes for up-to-date data

2. Scope

2.1 Included in This Release

Feature	Description
Step Sync (Android)	Health Connect integration with Google Fit, Samsung Health support
Step Sync (iOS)	Direct Apple Health integration
Activity Tracker Widget	Home page widget with multiple states (Not Connected, Solo Mode, Invite, Battle Mode)
Activity Challenge Page	Leaderboard, invite colleagues, view requests, update goal
Individual Challenges	Create, invite, accept/reject, compete with colleagues
Streak Tracking	Consecutive days meeting step goal
Background Sync	Automatic sync every 15 minutes
Timezone	IST (Indian Standard Time) only

2.2 Explicitly Out of Scope

Feature	Notes
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Calories tracking	Removed from scope
Distance tracking	Removed from scope
Water tracker	Separate feature
BMI / BP tracking	Separate feature
Push notifications for steps	Future phase
Streak milestone rewards	Admin-configurable in future
Manual step entry	Not supported

3. Step Sync Architecture

3.1 Android Architecture

On Android, Health Connect acts as a data bridge between third-party fitness apps and Habit Health. Health Connect does NOT track steps itself — it only aggregates data from connected apps.

Layer	Component	Description
1. Data Source	Phone sensors / Wearables	Actual step tracking hardware
2. Tracker App	Google Fit / Samsung Health / Fitbit	Records steps from sensors
3. Data Bridge	Health Connect	Aggregates data from tracker apps
4. Consumer	Habit Health App	Reads step data from Health Connect

3.2 iOS Architecture

On iOS, Apple Health is built-in and tracks steps automatically. Habit Health directly integrates with Apple Health — no additional tracker app is required.

Layer	Component	Description
1. Data Source	iPhone / Apple Watch	Built-in step tracking
2. Health Platform	Apple Health	Native health data storage
3. Consumer	Habit Health App	Reads step data from Apple Health

3.3 Supported Third-Party Apps (Android)

App	CTA	Behavior
Google Fit	Connect	Opens app if installed, else Play Store
Samsung Health	Connect	Opens app if installed, else Play Store
Other (Fitbit, Strava, etc.)	View Guide	Opens Step Sync Guide page

3.4 Compatibility Requirements

Platform	Minimum Version
Android	API Level 26 (Android 8.0) with Health Connect
iOS	iOS 8 and above with Apple Health

4. Connection Flow

4.1 Android Connection States

State	Condition	Banner Display	CTA
State 1	Health Connect permissions NOT granted	"Let's Get Moving" Allow sync with Health Connect	Allow
State 2	Health Connect OK, no tracker app connected	"Sync Step Tracking App" Allow Habit Health to read your steps	Connect
State 3	Sync error / failed refresh	"Couldn't Refresh the steps" Last sync: [timestamp]	Refresh
State 4	Successfully synced	"All caught up!" Last sync: [timestamp]	Refresh

4.2 iOS Connection States

iOS flow is simpler since Apple Health is built-in. Users only need to grant permissions — no third-party tracker app is required.

State	Condition	Banner Display	CTA
State 1	Apple Health permissions NOT granted	"Let's Get Moving" Allow sync with Apple Health	Allow
State 2	Sync error / failed refresh	"Couldn't Refresh the steps" Last sync: [timestamp]	Refresh
State 3	Successfully synced	"All caught up!" Last sync: [timestamp]	Refresh

4.3 Banner Color Coding

State	Background Color	Meaning
Needs Permission / Connect	Yellow/Cream	Action required
Sync Error	Orange	Warning - refresh needed
Success	Green	All good

4.4 Connection Flow Steps (Android)

1. User taps "Allow" → System opens Health Connect permissions screen
2. User grants "Allow All" permissions to Habit Health

3. System checks if tracker app (Google Fit/Samsung Health) is connected to Health Connect
4. If no tracker connected → Show "Connect" banner with tracker app options
5. User taps "Connect" → Opens modal with Google Fit, Samsung Health, Other Apps options
6. User selects app → Opens app (if installed) or Play Store
7. User connects tracker app to Health Connect
8. Return to Habit Health → Show "All caught up!" success banner
9. Background sync begins (every 15 minutes)

5. Activity Tracker Widget

The Activity Tracker widget appears on the Home Page in the "Join a challenge to unlock rewards" section. It displays different states based on the user's connection status and challenge participation.

5.1 Widget States

State	Badge	Display Content	CTA
Not Connected	Error icon	"Unable to sync the steps" 0 Steps / Sleepy character	Connect Steps
Solo Mode	"Solo Mode"	Today's steps, Goal, Last sync time, Running character	Take Challenge
Battle Mode (Upcoming)	"Battle Mode"	Same as Solo Mode + countdown timer to challenge start	View Status
Invite Received	"New challenge invite"	"[Name] Challenged you!" + Goal + Steps	View Invite
Battle Mode	"Battle Mode"	Today's steps, Mini leaderboard (top 3 ranks)	LeaderBoard

5.2 Widget State Priority

When multiple conditions could apply, the widget displays based on this priority order:

Priority	State	Condition
1 (Highest)	Invite Received	User has pending challenge invite
2	Not Connected	User accepted invite but steps not connected
3	Battle Mode	User is in an active challenge
4	Battle Mode (Upcoming)	Challenge created/joined but not yet started
5 (Default)	Solo Mode	Steps connected, no active challenge

Note: A user cannot receive new challenge invites while already in an active Individual Challenge.

5.3 Widget Data Display

Element	Display Format	Source
Steps	"1003 Steps"	Today's synced step count
Goal	"2.5k Goal"	User's daily step goal

Steps Taken	"1,030 Steps Taken"	Today's steps (same as above)
Last Sync	"Last sync: 2 minutes ago"	Relative time from last successful sync

5.4 Last Sync Time Display

Time Elapsed	Display Format
< 1 minute	"just now"
1-59 minutes	"X minutes ago"
1-23 hours	"X hours ago"
1-30 days	"X days ago"
> 30 days	"DD MMM YYYY" (e.g., "16 Sep 2025")

6. Activity Challenge Page

The Activity Challenge page (also called "StepUp Showdown") is the central hub for step tracking, challenges, and competing with colleagues. Users access this page by tapping the Activity Tracker widget.

6.1 Page Structure

- Header banner: "StepUp Showdown" with hero image
- Step Sync MFE banner (connection status - see Section 4)
- Description text and "Read more" link
- Action buttons: "Update Daily Goal" | "View Request" (with badge count)
- "Invite your Colleagues" CTA banner
- Content tabs: Leaderboard | Rules | FAQs

6.2 Update Daily Goal

Parameter	Value
Minimum goal	500 steps
Maximum goal	100,000 steps
Default goal	10,000 steps
Can change during Individual Challenge?	No
Can change during CUG Challenge (Stepathon)?	Yes

6.3 View Request

Shows pending challenge invites received from colleagues. Badge shows count of pending requests. Tapping opens the invite details where user can Accept or Reject.

6.4 Invite Your Colleagues

Feature	Specification
Who can be invited	All users in the same CUG
Search by	Name, Email only
Invite limit	No limit
Can invite user in active challenge?	No — user must not be in an active Individual Challenge

7. Individual Challenge Flow

7.1 Challenge Creation

Users create Individual Challenges from the Activity Challenge page by clicking "Invite your Colleagues".

Parameter	Options	Default
Step Goal	500 - 100,000 steps	User's current daily goal
Duration	7, 14, 21, 28 days	21 days
Start Date	T+1 to T+7 (tomorrow to 7 days from today)	T+2 (2 days from today)
Invites Required	At least 1 colleague	Cannot start without invites

7.2 Invite Flow

1. Creator searches for colleagues by Name or Email
2. Creator selects colleagues and sends invites
3. Invitees see "New challenge invite" on their Activity Tracker widget
4. Invitee taps "View Invite" → Opens invite details modal
5. Modal shows: Challenge goal, Duration, Start date, Creator name
6. Invitee taps "Accept" or "Reject"
7. On Accept: Invitee joins challenge, widget shows countdown to start
8. On Reject: Invite removed, creator notified

7.3 Invite Rules

Rule	Behavior
User in active Individual Challenge	Cannot receive new invites
Overlapping challenge dates	Cannot receive invites for overlapping dates
Rejection limit	After 3 rejections from same person → blocked for 30 days
Request expiration	Invites expire once start date passes
Creator can withdraw	Yes, before invitee accepts
Multiple invites to same person	Show most recent inviter's name on widget

7.4 During Challenge

- Challenge starts automatically at midnight IST on the start date
- All participants' daily goals are updated to the challenge goal
- Widget shows "Battle Mode" with mini leaderboard (top 3)
- Leaderboard is based on streak points (days meeting goal)
- Tiebreaker: Higher total step count wins
- Users can view full leaderboard on Activity Challenge page
- Background sync refreshes leaderboard every 15 minutes

7.5 Challenge Completion

- Challenge ends at 11:59 PM IST on the end date
- 24-hour buffer period for final step sync
- Final leaderboard announced after buffer period
- User's daily goal is restored to their pre-challenge goal
- Widget returns to "Solo Mode"
- Past challenges accessible from "All Challenges" section

7.6 Scoring System

Metric	Description
Primary Score	Streak points — each day meeting step goal = 1 point
Tiebreaker	Total step count — higher wins
Calculation Start	From the date user joined the challenge
Winner	User with most streak points (or highest steps if tied)

8. Streak System

8.1 Streak Definition

A streak is the count of consecutive days where the user meets their daily step goal. Streaks encourage daily engagement and build healthy habits.

8.2 Streak Rules

Rule	Behavior
Streak increment	User meets daily step goal by 11:59 PM IST
Streak break	User does NOT meet goal by end of grace period
Grace period	2 hours after midnight (until 2:00 AM IST)
Late sync restoration	If steps sync after grace period but were actually walked, streak is restored retroactively
Goal change impact	Changing goal does not break streak — based on goal at 11:59 PM
Streak display	Current streak only (not longest ever)

8.3 Streak Update Timing

Streaks are calculated at approximately 11:59 PM IST each day, with a 2-hour grace period to account for sync delays. If a user's steps are updated in Apple Health/Health Connect before the grace period ends, the streak will be maintained.

8.4 Retroactive Streak Restoration

If a user's steps sync late (e.g., at 3 AM) but they actually achieved their goal the previous day, the system will restore their streak once the data syncs. This prevents penalizing users for sync delays beyond their control.

8.5 Streak Rewards

Streak milestone rewards are not included in the current release. However, the system will support admin-configurable streak rewards in a future phase (e.g., badges or Care Coins for 7-day, 30-day, 100-day streaks).

9. Leaderboard

9.1 Leaderboard Visibility

Context	Who is Visible
Before challenge starts	All CUG users who have connected step sync
During challenge	Toggle: All CUG users Only challenge participants ("Yours")
After challenge	Challenge participants only (final standings)

9.2 Leaderboard Columns

Column	Displayed	Notes
Rank	Yes	Position based on filter criteria
User Name	Yes	Display name
Steps	Yes	Based on selected filter (today/week/month/overall)
Current Streak	Yes	Number of consecutive days meeting goal
Daily Goal	No	Other users' goals are not shown

9.3 Leaderboard Filters

Filter	Steps Shown	Ranking By
Today	Today's steps	Today's step count
Week	This week's total (Mon-Sun)	Weekly total
Month	This month's total	Monthly total
Overall	All-time total	All-time total

9.4 Leaderboard Refresh

Trigger	Behavior
Page open	Fetch latest data immediately
While on page	Auto-refresh every 15 minutes
Manual refresh	User taps "Refresh" button

9.5 Empty Leaderboard States

Scenario	Display Message
Challenge not started	"Leaderboard will be updated once challenge starts"
No users with step sync	"No users have connected step sync yet"
Filter shows no data	"No step data for this period"

10. Technical Requirements

10.1 Background Sync

Parameter	Value
Sync frequency	Every 15 minutes
Trigger pages	Home Page, Play Page, Challenge Listing Page
Sync condition	Health Connect/Apple Health connected AND tracker app connected (Android)
Error handling	Show last synced data + toast "Couldn't refresh steps, will retry"

10.2 Data Storage

Data Type	Retention Period
Daily step data	1 year
Streak history	1 year
Challenge history	1 year
Leaderboard snapshots	Duration of challenge + 30 days

10.3 Device Management

Scenario	Behavior
User switches device (Android to iOS)	Must reconnect on new device, warning popup shown
User logs in on second device	Warning: "This account is connected with another device"
Multiple devices same day	Use latest synced device, max 1 device change per day
Steps from multiple sources	Health Connect/Apple Health handles deduplication

10.4 Timezone

All step tracking, streak calculations, and challenge timings use IST (Indian Standard Time). Multi-timezone support is not included in this release.

10.5 Admin Configuration

Setting	Description
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Default step goal for CUG	Admin can set default goal for all users in CUG
Enable/disable step sync	Admin can enable or disable step sync feature per CUG
Step sync scoping	Admin enables step sync for specific CUGs

11. Edge Cases & Error Handling

Edge Case	Expected Behavior
User has 0 steps for a day	Display "0" — no special handling
Sync fails repeatedly	Show last synced data with error banner and timestamp
User revokes Health Connect permission	Return to State 1 (permission required)
User uninstalls tracker app	Return to State 2 (tracker app required)
Challenge invite expires	Remove from pending invites list
User quits challenge before start	Allowed — user removed from participants
User quits challenge after start	Allowed — user ineligible for rewards
All invitees reject	Challenge auto-cancelled when start date passes
Late step sync (after grace period)	Streak restored retroactively
Device storage full	Show error message, suggest clearing space
Network connection lost	Use cached data, retry sync when online
Few users have synced steps (leaderboard)	Show available users; empty state if none

12. Acceptance Criteria

12.1 Step Sync Connection

- User can connect to Health Connect (Android) or Apple Health (iOS)
- User is guided to install/connect tracker app if needed (Android)
- Sync banner shows correct state (Allow → Connect → Success)
- Error state shows with last sync timestamp when sync fails
- Manual "Refresh" button triggers immediate sync attempt

12.2 Activity Tracker Widget

- Widget shows correct state based on priority (Invite > Not Connected > Battle > Battle Upcoming > Solo)
- Widget displays today's steps accurately
- Widget displays user's daily goal
- Last sync time updates correctly (relative format)
- Widget CTA navigates to Activity Challenge page
- Battle Mode (Upcoming) shows countdown timer and 'View Status' CTA

12.3 Activity Challenge Page

- Page shows Step Sync MFE banner with correct connection state
- "Update Daily Goal" allows changing goal within limits (500-100,000)
- "View Request" shows pending invites with badge count
- "Invite your Colleagues" opens search modal
- Search works by Name and Email
- Leaderboard tabs (Leaderboard | Rules | FAQs) function correctly
- Leaderboard filters (Today | Week | Month | Overall) display correct data

12.4 Individual Challenge

- User can create challenge with goal, duration, and start date
- User can search and invite colleagues
- Invitee receives notification and can Accept/Reject
- Challenge starts automatically at midnight on start date
- Leaderboard updates during challenge based on streak points

- Challenge ends at 11:59 PM on end date with 24-hour buffer
- User's goal is restored after challenge ends
- Past challenges are accessible in "All Challenges"

12.5 Streak System

- Streak increments when user meets daily goal by 11:59 PM + 2hr grace
- Streak breaks when goal is not met after grace period
- Streak is restored retroactively if late sync shows goal was met
- Current streak is displayed on leaderboard

12.6 Technical

- Background sync runs every 15 minutes on trigger pages
- Step data is retained for 1 year
- Device switch shows warning and requires reconnection
- Admin can configure default goal and enable/disable sync per CUG

— End of Document —